Numerics for ordinary differential equations

University of Milan, Italy

Biomatematica 2

The model problem

Consider the Cauchy problem

$$\begin{cases} \dfrac{dy(t)}{dt} = f(t, y(t)) \quad t \in (t_0, T) \\ y(t_0) = y_0. \end{cases}$$

Suppose that:

$f = f(t, y) : (t_0, T) \times \mathbb{R} \to \mathbb{R}$ is Lipschitz with respect to y, uniformly with respect to t, i.e.

$\exists\, L > 0$ such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2| \qquad \forall t \in (t_0, T),\ \forall y_1, y_2 \in \mathbb{R}.$$

Numerical approximation

- Introduce $N+1$ discretization nodes in $[t_0, T]$:

$$h > 0, \quad t_j = t_0 + jh, \quad j = 0, 1, ..., N, \quad t_N \leq T.$$

- Denote by y_j the exact solution $y(\cdot)$ evaluated in the node t_j, i.e. $y_j := y(t_j)$, $j = 0, 1, ..., N$.

Definition. A numerical method for the approximation of the Cauchy problem is an algorithm that constructs $N+1$ real values u_j, that approximate y_j, $\forall j = 0, 1, ..., N$, i.e. $u_j \approx y_j$.

Basic definitions

Definition. A numerical method for the approximation of the Cauchy problem is called one-step method if, $\forall n \geq 0$, u_{n+1} depends only on the previous step u_n and not on u_j with $j < n$. Otherwise it is called multistep.

Definition. A numerical method for the approximation of the Cauchy problem is called explicit if, $\forall n \geq 0$, u_{n+1} is computed as a function of the previous steps u_j with $j \leq n$. It is called instead implicit if, $\forall n \geq 0$, u_{n+1} depends implicitly on itself through the function f.

Euler methods

- Explicit (or Forward) Euler method. Set $u_0 = y_0$. Then $\forall n \geq 0$
$$u_{n+1} = u_n + h\ f(t_n, u_n).$$

- Implicit (or Backward) Euler method. Set $u_0 = y_0$. Then $\forall n \geq 0$
$$u_{n+1} = u_n + h\ f(t_{n+1}, u_{n+1}).$$

Crank-Nicolson and Heun methods

- Crank-Nicolson method. Set $u_0 = y_0$. Then $\forall n \geq 0$

$$u_{n+1} = u_n + \frac{h}{2}\left(f(t_n, u_n) + f(t_{n+1}, u_{n+1})\right).$$

- Heun method. Set $u_0 = y_0$. Then $\forall n \geq 0$

$$u_{n+1} = u_n + \frac{h}{2}\left(f(t_n, u_n) + f(t_{n+1}, u_n + h\ f(t_n, u_n))\right).$$

BDF (Backward Differentiation Formula) methods

- BDF2. Given $u_0 = y_0$ and u_1, then $\forall n \geq 1$

$$u_{n+1} = \frac{4}{3}u_n - \frac{1}{3}u_{n-1} + \frac{2}{3}h\ f(t_{n+1}, u_{n+1}).$$

- BDF3. Given $u_0 = y_0$, u_1 and u_2, then $\forall n \geq 2$

$$u_{n+1} = \frac{18}{11}u_n - \frac{9}{11}u_{n-1} + \frac{2}{11}u_{n-2} + \frac{6}{11}h\ f(t_{n+1}, u_{n+1}).$$

Consistency I

Consider the general form of an explicit one-step method:

$$u_{n+1} = u_n + h\ \Phi(t_n, u_n, f(t_n, u_n), h),$$

where the function Φ is called incremental function.

Being $y(\cdot)$ the exact solution of the Cauchy problem, set

$$\epsilon_{n+1} = y_{n+1} - y_n - h\ \Phi(t_n, y_n, f(t_n, y_n), h) \quad 0 \leq n \leq N-1.$$

Thus ϵ_{n+1} is the error we commit by claiming that the exact solution satisfies the numerical scheme.

Consistency II

Definition. We call local truncation error the quantity

$$\tau_{n+1}(h) = \frac{\epsilon_{n+1}}{h}.$$

We call global truncation error the quantity

$$\tau(h) = \max_{0 \le n \le N-1} \tau_{n+1}(h).$$

Definition. A numerical method is called consistent if

$$\lim_{h \to 0} \tau(h) = 0.$$

A numerical method is called consistent with order p if

$$\tau(h) = O(h^p).$$

Zero-stability and convergence

Definition. A numerical method is called zero-stable if, in a bounded interval (t_0, T), small perturbations on the data yield small perturbations on the approximate solution, for $h \to 0$.

Definition. A numerical method is called convergent with order p, if

$$\exists C > 0 \ : \qquad |u_n - y_n| \leq Ch^p \qquad \forall n : \ 0 \leq n \leq N.$$

Theorem. A numerical method is convergent if and only if it is consistent and zero-stable.

Absolute stability

Consider the model problem

$$\begin{cases} \dfrac{dy(t)}{dt} = -\lambda y(t) \quad t \in (0, +\infty) \quad \lambda > 0 \\ y(0) = 1 \end{cases} \tag{1}$$

with exact solution $y(t) = e^{-\lambda t}$.

Definition. A numerical method is called absolutely stable if, when applied to the model problem (1),

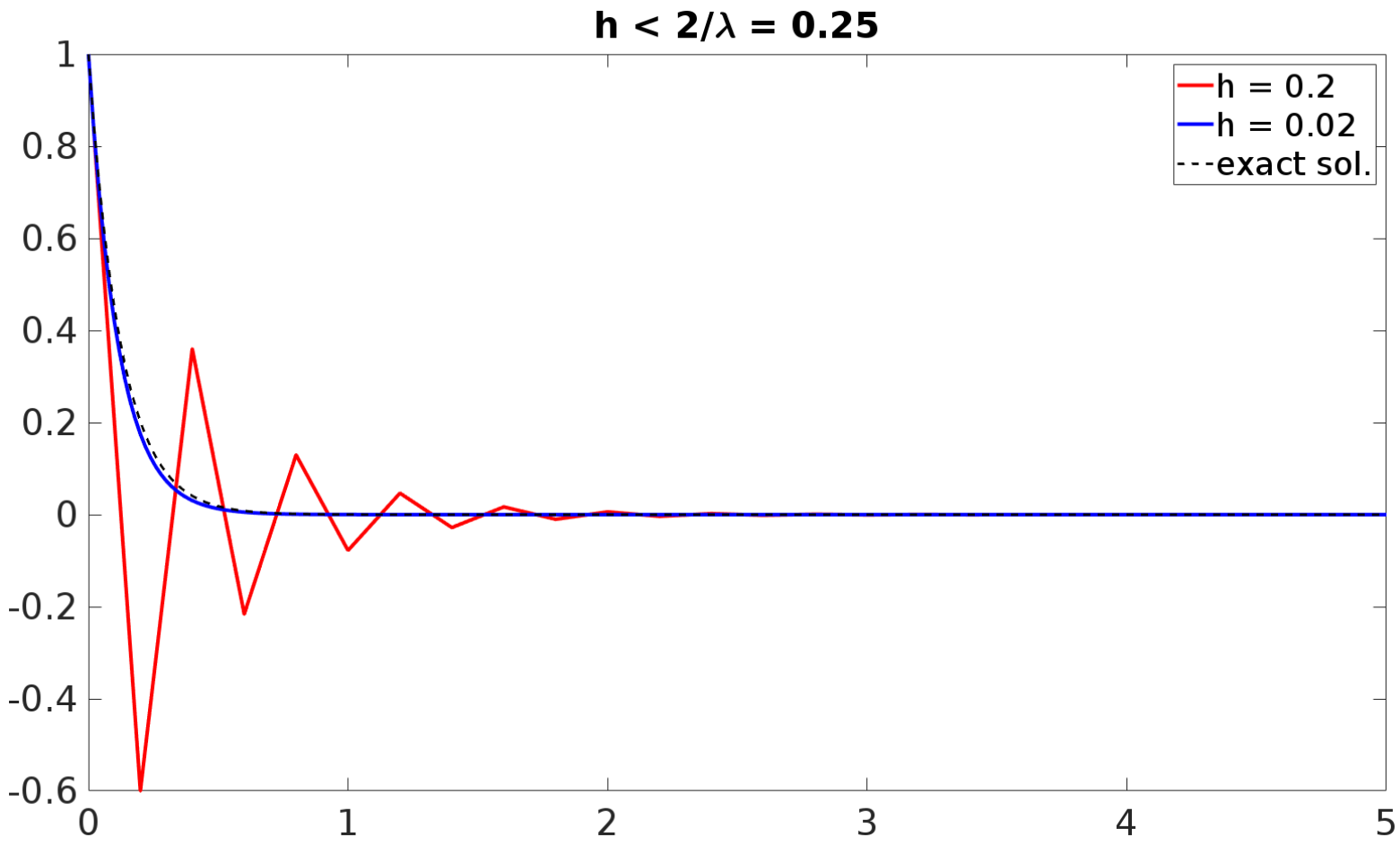
$$u_n \to 0 \quad \text{for} \quad t_n \to +\infty.$$

Absolute stability for one-step methods

- Explicit Euler is absolutely stable if and only if $h < \frac{2}{\lambda}$.
- Implicit Euler is unconditionally absolutely stable.
- Heun is absolutely stable if and only if $h < \frac{2}{\lambda}$.
- Crank-Nicolson is unconditionally absolutely stable.

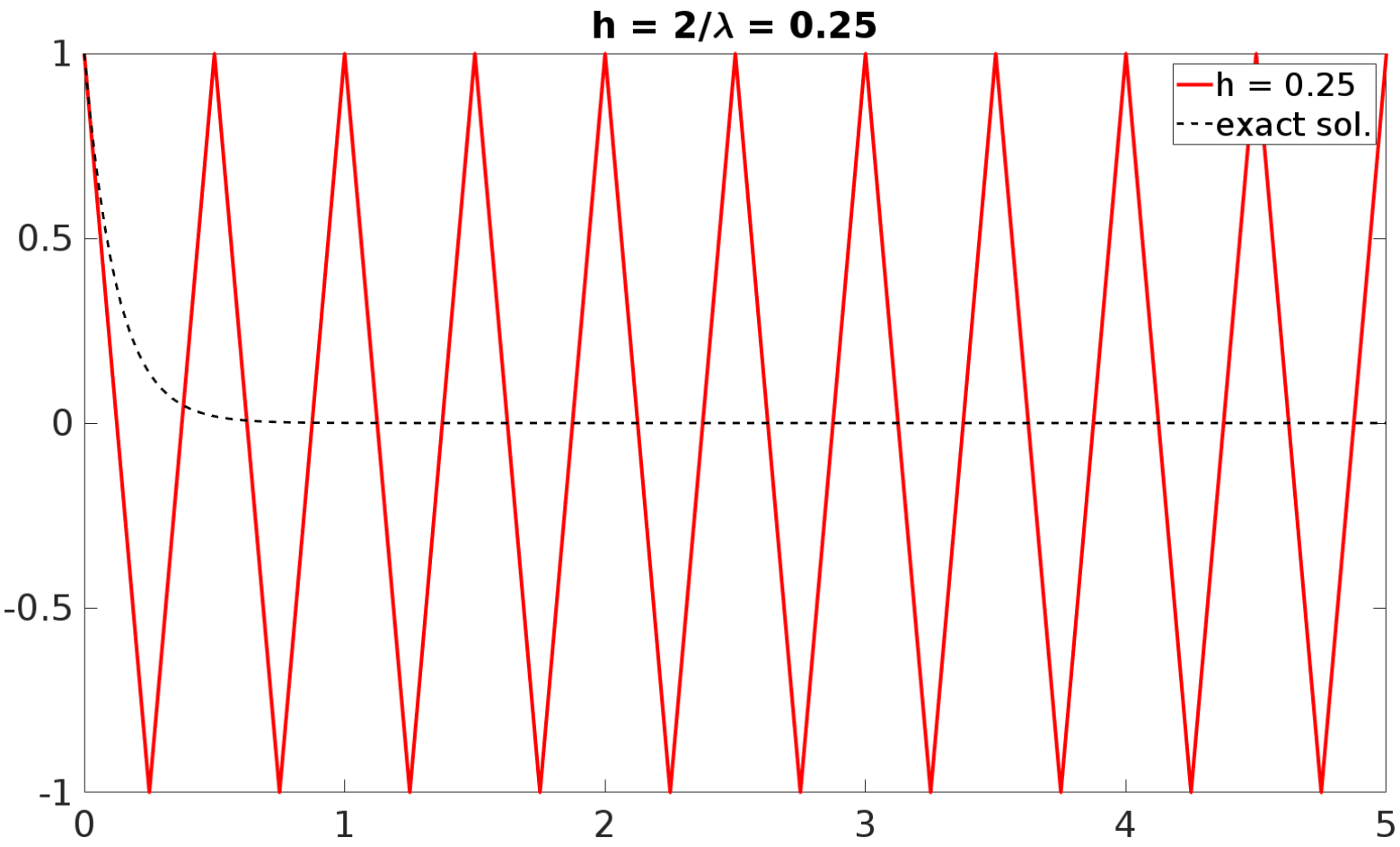
Example 1: Explicit Euler with $h < \frac{2}{\lambda}$

$\lambda = \mathbf{8}$



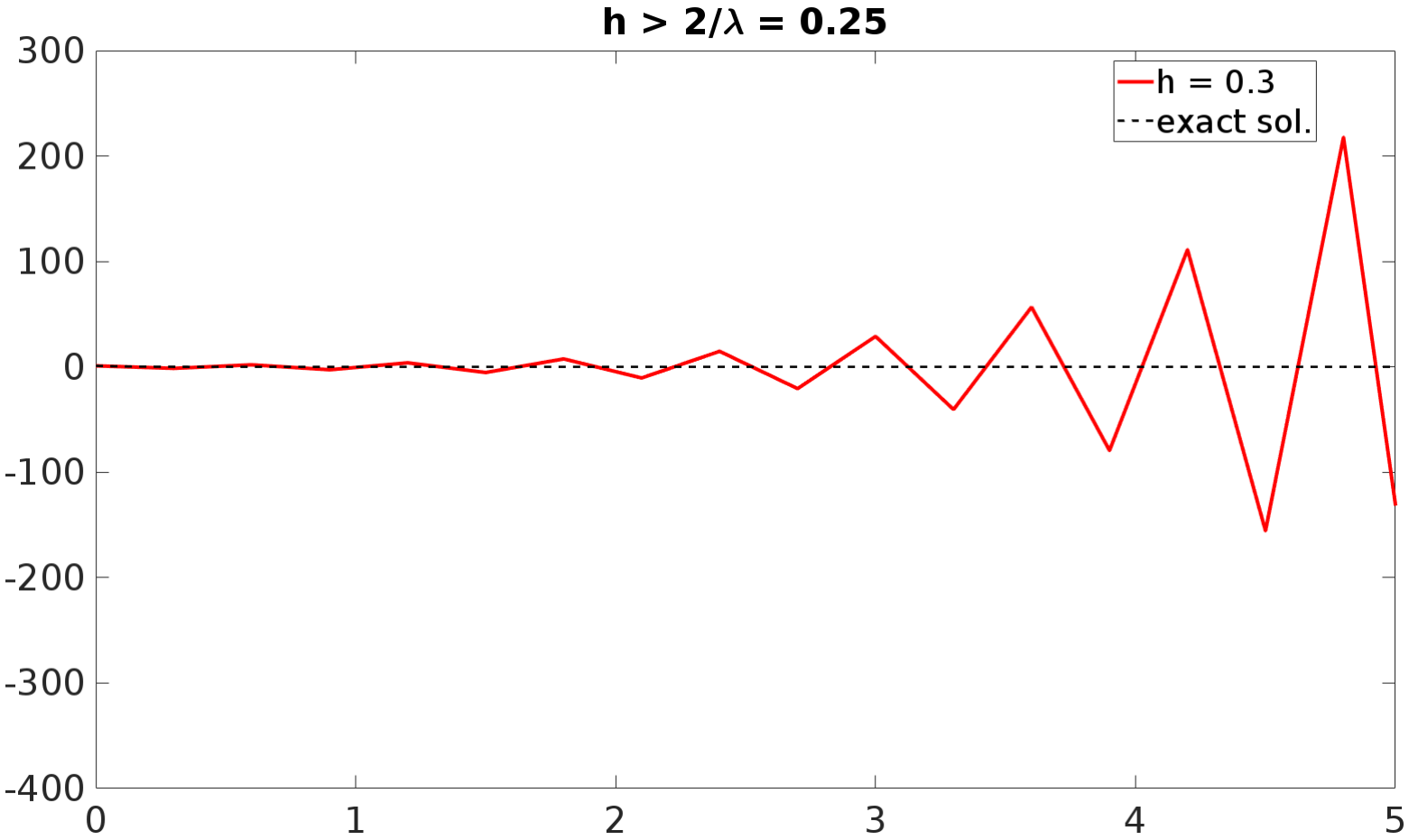
Example 2: Explicit Euler with $h = \frac{2}{\lambda}$

$\lambda = \mathbf{8}$



Example 3: Explicit Euler with $h > \frac{2}{\lambda}$

$\lambda = \mathbf{8}$



The Matlab ODE functions

Matlab provides a set of functions in the ODE suite, which solve ordinary differential equations, for example the **ode23**, **ode45**, **ode15s** functions.

Sintax: [T,Y] = odeXX(fun,Tspan,y0)

Input:

- fun = $f(t, y)$ given right hand side function
- Tspan = $[t_0, t_{max}]$ integration interval
- y0 = initial condition

Output:

- T = $[t_0, t_1, ..., t_N]^T$ vector of discretization nodes
- Y = $[Y_0, Y_1, ..., Y_N]^T$ solution vector

Example 1: a scalar ODE

Solve with the **ode23**, **ode45** and **ode15s** functions the following ODE:

$$\begin{cases} \dfrac{dy}{dt} = -4\,(y+1)\,\sin(t)\,\cos(t), \quad 0 \le t \le 2 \\ y(0) = e - 1, \end{cases}$$

whose exact solution is

$$y(t) = e^{\cos(2\,t)} - 1.$$

Compare the errors of the three methods in the ∞-norm.

Example 2: a system of two ODEs

Solve with the **ode45** function the following predator-prey system on the interval $[0, 200]$:

$$\begin{cases} \dfrac{dy_1}{dt} = -a\, y_1 + \dfrac{b\, y_2}{c + ky_2} y_1 & y_1(0) = 0.5 \\ \dfrac{dy_2}{dt} = (d - e\, y_2) y_2 - \dfrac{f\, y_2}{c + ky_2} y_1 & y_2(0) = 1, \end{cases}$$

where

- $a = 0.5$;
- $b = d = e = f = 1$;
- $c = 0.3$;
- $k = 0.7$.